CurriculumAI: Technical Evaluation Report

1. Technical Evaluation Overview

Design and Evaluation of a Retrieval-Augmented Question Answering System for Computer Programming Students, BSc Computer Science, De Montfort University.

This report analyses the technical-evaluation workbook.

All five worksheets were examined. Numerical results were independently recalculated from the 48 response records and checked against the summary formulas. Qualitative analysis uses the complete recorded answers and their evidence notes. Selected source passages were checked directly in RAG_Knowledge_source.zip. External programming documentation was used only to clarify technical issues, never to expand the corpus used for unsupported-claim scoring.

The dataset contains 24 matched question pairs: Q01-Q16 are labelled fully supported, Q17-Q20 partially supported and Q21-Q24 unsupported. All 48 response statuses are Answered, and all capture dates are recorded as 7 November 2026. The guide specifies identical questions, fresh single-turn conversations and retention of the first response without selective regeneration.

CurriculumAI produced 16 recorded Tier-1 responses, Q01-Q16, and eight Tier-2 responses, Q17-Q24. Tier 1 displays course-source citations. Tier 2 provides external OpenAI API generation accompanied by an explicit notice that sufficient course evidence was not found. It is not the ChatGPT application.

The response-level configuration entries are GPT-6 Luna (OpenAI API) for CurriculumAI and ChatGPT Free - standard mode; Think disabled for ChatGPT.

The five measures answer different questions: factual accuracy concerns technical correctness; relevance concerns answering the intended Scala question; citation faithfulness concerns actual claim-to-citation support; transparency concerns disclosed information origin; unsupported-claim rate concerns support anywhere within the fixed evaluation corpus. Missing citations do not establish factual error or absent corpus support.

2. Verified Numerical Results

Answer-quality scores

Table 1. Answer-quality score distributions and applicable means.

Metric	System	Score 0	Score 1	Score 2	N/A	Applicable n	Mean / 2
Factual accuracy	CurriculumAI	0	0	24	0	24	2.000
Factual accuracy	ChatGPT	1	1	22	0	24	1.875
Relevance	CurriculumAI	0	0	24	0	24	2.000
Relevance	ChatGPT	1	4	19	0	24	1.750
Citation faithfulness	CurriculumAI	0	0	16	8	16	2.000
Citation faithfulness	ChatGPT	0	0	0	24	0	Not applicable
Source transparency	CurriculumAI	0	0	24	0	24	2.000
Source transparency	ChatGPT	24	0	0	0	24	0.000

There are no blank values among these recorded metric entries. N/A is excluded from numeric means, rather than converted to zero.

Accuracy: CurriculumAI received full marks on all questions. ChatGPT received 0 on Q05, where Python generator behaviour replaced the Scala construct, and 1 on Q21, where the answer incorrectly restricted a C enum variable to named enumerators. The remaining 22 ChatGPT answers received 2. The paired scores differ on only two questions; the other 22 pairs are tied. Thus the mean difference of 0.125 points is concentrated rather than evidence of a broad accuracy advantage across every topic.

Relevance: ChatGPT received 0 on Q05 and 1 on Q07, Q08, Q09 and Q14. The concerns are, respectively, an incorrect construct, JavaScript-specific filtering, a C# loop rather than a Scala collection method, JavaScript-specific flattening, and unrelated meanings of “class”. The other 19 pairs are tied at 2. Foreign-language examples did not automatically attract a deduction: Q03, Q06, Q10-Q12, Q15 and Q24 retain full relevance.

Citations: Sixteen CurriculumAI responses contain actual citations, and all 16 received faithfulness 2. The eight Tier-2 provenance notices are not citations and receive N/A. None of the 24 ChatGPT responses contains an actual citation. Therefore there is no assessable ChatGPT citation-faithfulness mean and no evidence of unfaithful ChatGPT citations in this dataset. The numerical comparison is citation availability, 16/24 versus 0/24, followed by conditional faithfulness within the cited subset.

The recorded source checks connect the cited claims to actual slide text or examples. Important qualifications are that Q02 cites a DOCX block associating val with constants, and Q03 cites a worked head example rather than a standalone definition. Q08 and Q12 rely on combined evidence from multiple cited locations. The notes record screenshot resolution of the missing Q01 source marker and malformed Q13 bracket in the workbook transcription. These are inferential or transcription qualifications, not recorded scores of partial support.

Transparency: CurriculumAI's 24 full scores comprise 16 identifiable course-source displays and eight explicit external-generation notices. ChatGPT's 24 zero scores indicate absence of these answer-specific disclosures. This measure does not show that ChatGPT was deceptive, nor that every transparent CurriculumAI claim was course-supported. An external-generation notice earns transparency credit while its answer can remain largely unsupported by the corpus.

Unsupported-claim results

Pooled rate = total unsupported claims / total assessable claims. Mean response rate = the arithmetic mean of individual response percentages. All 48 responses have non-zero assessable counts, so no N/A or unresolved records are excluded here.

Table 2. Unsupported claims by response tier and system.

System / group	Responses	Supported	Unsupported U	Total N	Pooled rate	Mean response rate
CurriculumAI - Tier 1	16	40	0	40	0.00%	0.00%
CurriculumAI - Tier 2	8	13	27	40	67.50%	71.40%
CurriculumAI - Overall	24	53	27	80	33.75%	23.80%
ChatGPT - Overall	24	86	103	189	54.50%	50.73%

CurriculumAI's overall pooled rate is 20.75 percentage points lower. Its 27/80 calculation uses the actual combined claim counts, not an average of the tier percentages. Longer answers contribute more claims to the pooled measure, whereas each answer receives equal weight in the response-level mean.

Table 3. Pooled unsupported-claim rates by matched question group.

Matched question group	CurriculumAI supported	CurriculumAI U/N	Pooled rate	ChatGPT supported	ChatGPT U/N	Pooled rate
Q01-Q16: Fully supported	40	0/40	0.00%	57	47/104	45.19%
Q17-Q20: Partially supported	12	14/26	53.85%	25	23/48	47.92%
Q21-Q24: Unsupported	1	13/14	92.86%	4	33/37	89.19%

The overall advantage does not hold within the latter two strata: ChatGPT has lower pooled unsupported-claim rates in both. “Unsupported question” describes the expected answer points; it does not prevent a response adding a corpus-supported statement, such as the two Boolean values in Q22.

Numerical audit: Recalculated distributions, applicable counts, means, tier totals and question-group pooled rates agree with Evaluation Summary. Its one-decimal percentages are display rounding. No current formula errors or #REF! references were found.

3. Question-by-Question Comparative Results

Notation: A/R/F/T = factual accuracy / relevance / citation faithfulness / source transparency, each scored 0-2. NA means no assessable citation. U/N and rates reproduce the saved claim decisions; percentages are rounded only for display.

Table 4. Question-by-question scores and unsupported-claim rates.

Question	CurriculumAI A/R/F/T	CurriculumAI U/N; rate	ChatGPT A/R/F/T	ChatGPT U/N; rate
Q01	2/2/2/2	0/1; 0.0%	2/2/NA/0	0/1; 0.0%
Q02	2/2/2/2	0/1; 0.0%	2/2/NA/0	0/1; 0.0%
Q03	2/2/2/2	0/1; 0.0%	2/2/NA/0	1/2; 50.0%
Q04	2/2/2/2	0/1; 0.0%	2/2/NA/0	0/4; 0.0%
Q05	2/2/2/2	0/3; 0.0%	0/0/NA/0	6/6; 100.0%
Q06	2/2/2/2	0/2; 0.0%	2/2/NA/0	1/8; 12.5%
Q07	2/2/2/2	0/2; 0.0%	2/1/NA/0	2/6; 33.3%
Q08	2/2/2/2	0/5; 0.0%	2/1/NA/0	3/5; 60.0%
Q09	2/2/2/2	0/2; 0.0%	2/1/NA/0	4/7; 57.1%
Q10	2/2/2/2	0/3; 0.0%	2/2/NA/0	3/7; 42.9%
Q11	2/2/2/2	0/2; 0.0%	2/2/NA/0	3/8; 37.5%
Q12	2/2/2/2	0/6; 0.0%	2/2/NA/0	5/9; 55.6%
Q13	2/2/2/2	0/2; 0.0%	2/2/NA/0	2/8; 25.0%
Q14	2/2/2/2	0/2; 0.0%	2/1/NA/0	5/8; 62.5%
Q15	2/2/2/2	0/1; 0.0%	2/2/NA/0	1/1; 100.0%
Q16	2/2/2/2	0/6; 0.0%	2/2/NA/0	11/23; 47.8%
Q17	2/2/NA/2	4/6; 66.7%	2/2/NA/0	6/10; 60.0%
Q18	2/2/NA/2	3/6; 50.0%	2/2/NA/0	8/18; 44.4%
Q19	2/2/NA/2	6/11; 54.5%	2/2/NA/0	8/15; 53.3%
Q20	2/2/NA/2	1/3; 33.3%	2/2/NA/0	1/5; 20.0%
Q21	2/2/NA/2	3/3; 100.0%	1/2/NA/0	7/7; 100.0%
Q22	2/2/NA/2	2/3; 66.7%	2/2/NA/0	5/9; 55.6%
Q23	2/2/NA/2	5/5; 100.0%	2/2/NA/0	13/13; 100.0%
Q24	2/2/NA/2	3/3; 100.0%	2/2/NA/0	8/8; 100.0%

4. Major Findings and Representative Answer Examples

Q01 - Equivalent factual performance

Exact question: "In the Scala environment, which runtime executes compiled Scala bytecode?"

Both answers state: "The Java Virtual Machine (JVM) executes compiled Scala bytecode." CurriculumAI adds [S3]; ChatGPT supplies no citation. Both receive accuracy 2, relevance 2 and 0/1 unsupported claims. Class_Libraries_Packages_and_Testing_Week25part2_Block-C.pptx, slide 10, substantiates execution of compiled class files on the JVM.

Observation: The substantive answers are equivalent. Interpretation: This is evidence of an inspectability difference, not a correctness difference. ChatGPT's uncited answer is nevertheless fully supported by the corpus.

Q05 - A different meaning of yield

Exact question: "What does yield produce in a for expression?"

CurriculumAI says the iterated values "form a single collection". ChatGPT begins: "In a Python for expression, yield produces a value from a generator," then describes suspension and resumption. The lecture's intended construct is Scala collection construction, supported by For-Expressions_Week22part1_Block-B.pptx, slides 13-14. The conceptual distinction is confirmed by the Scala control-structures documentation and Python yield-expression documentation.

Observation: The scores are accuracy/relevance 2/2 versus 0/0, and unsupported counts 0/3 versus 6/6. Interpretation: The answer fails the intended Scala question; this does not mean every statement about Python generators is false. The accuracy rubric is being applied to correctness as an answer to the intended construct. That boundary should be explicit to avoid interpreting the same contextual failure as two independent kinds of evidence. Flattened code indentation in the workbook is documented as a transcription issue, not a generated Python syntax error.

Q07 - Transferable filtering, incomplete Scala framing

Exact question: "What does filter return?"

CurriculumAI describes a new collection containing elements satisfying a predicate, including a new list when filtering a list. ChatGPT says: "The filter() function returns a new array containing only the elements that pass a specified condition," followed by JavaScript code and "truthy" callback results.

Observation: Both receive accuracy 2, but relevance is 2 versus 1. Unsupported counts are 0/2 versus 2/6. The selection principle transfers to the course concept; the JavaScript example and truthiness assertion are classified outside the corpus. Applying-Functions-over-Collections_Week22part2_Block-B.pptx, slide 10, supports new-collection predicate selection.

Interpretation: ChatGPT provides useful conceptual overlap, rather than a wholly irrelevant answer. Its array-specific contract is not adequately tied to the intended Scala collection.

Q08 - Collection method versus language statement

Exact question: "What does foreach do?"

CurriculumAI says it "takes a function and applies it to each element of a collection" and explains Unit and side effects. ChatGPT explains a C# foreach loop with a body printing fruit names. Both address iteration, but only CurriculumAI establishes the taught function-accepting method contract.

Observation: Accuracy is 2 for both; relevance is 2 versus 1; unsupported counts are 0/5 versus 3/5. The source is Applying-Functions-over-Collections_Week22part2_Block-B.pptx, slides 6-7.

Interpretation: The distinction concerns the construct the student is learning, not an alleged error in the C# example. Neither answer's pedagogical effectiveness was experimentally measured.

Q09 - A substantively transferable explanation

Exact question: "What does flatMap do?"

CurriculumAI describes returning collections from a mapping function and concatenating them. ChatGPT begins: "flatMap() is a JavaScript array method," but accurately illustrates mapping followed by flattening, including numerical and hobby-list examples.

Observation: Both receive accuracy 2; relevance is 2 versus 1. Unsupported counts are 0/2 versus 4/7. Slide 16 of the collections lecture supports the shared conceptual explanation.

Interpretation: This is a borderline relevance difference. ChatGPT's substantive explanation is transferable, and the deduction depends on requiring an explicitly Scala collection-method account. The saved score is retained here.

Q14 - Resolving an ambiguous short question

Exact question: "What is a class?"

CurriculumAI answers: "A class is a blueprint for modelling objects." ChatGPT includes that programming meaning but also school classes, social classes and general categories, then asks which sense was intended.

Observation: Accuracy remains 2 for both; relevance is 2 versus 1. Unsupported counts are 0/2 versus 5/8. Classes_Week25part1_Block-C.pptx, slide 3, supports the programming definition.

Interpretation: CurriculumAI supplies a more focused response in this case. However, the exact question contains no explicit Scala or programming qualifier, and ChatGPT was not supplied the course corpus. The observation cannot isolate retrieval as the cause; application context and question ambiguity are competing explanations. The everyday meanings are counted as corpus-unsupported statements, not necessarily false statements.

Q15 - Correct keyword, 100% corpus-unsupported

Exact question: "Which keyword creates an instance of a class?"

Both supply new. ChatGPT explicitly says "in Java" and shows Car myCar = new Car();. Both receive accuracy and relevance 2. Nevertheless, its sole segmented claim receives unsupported status, producing 1/1 = 100%, compared with CurriculumAI's 0/1.

Observation: The claim notes require evidence for the Java-specific assertion; Classes_Week25part1_Block-C.pptx, slide 3, demonstrates Scala instantiation. Interpretation: The recorded decision is defensible under a strict language-specific substantiation rule, but it is highly sensitive to segmentation. Treating the shared new concept separately from the Java qualification could produce a different rate. The factual-accuracy notes also use the Scala slide to corroborate the core keyword answer, highlighting different support thresholds across measures. This must not be described as a wholly incorrect answer.

Q16 - Additional useful detail increases corpus exposure

Exact question: "What is JUnit?"

CurriculumAI describes a framework for automated Scala/Java tests and expected-versus-actual output comparison. ChatGPT correctly identifies JUnit as a Java testing framework, then supplies a Jupiter example, lifecycle annotations, benefits and JUnit 4/5 comparisons.

Observation: Both receive accuracy and relevance 2. CurriculumAI has 0/6 unsupported claims; ChatGPT has 11/23. The corpus supports JUnit automation and comparison in Unit_Testing_and_JUnit_Week23part3_Block-C.pptx, slides 23-25. DOCX exercises additionally support @Before, so it was not classified unsupported merely because it was absent from the cited introductory slides.

Interpretation: ChatGPT's longer answer contains 12 supported claims, twice CurriculumAI's six, as well as additional unsupported material. Its breadth can be informative without satisfying corpus-boundedness. The rate measures the latter, not overall educational value.

Q18 - Complete answers without an unrequested detail

Exact question: "What is a Scala List, and what does it mean for List to be covariant?"

CurriculumAI explains an immutable linked sequence and List[+A]. ChatGPT explains an immutable ordered collection, subtype preservation, assignments and a longer covariance rationale. Its key explanation is: "a List of a subtype can be used wherever a List of its supertype is expected."

Observation: Both receive accuracy and relevance 2. ChatGPT's omission of the linked representation is not treated as failure to answer the explicit conceptual question. Arrays_Week21part3_Block-B.pptx, slide 4, supports immutability and linked representation, but the covariance explanation is classified outside the corpus. The rates are 3/6 = 50.0% for CurriculumAI and 8/18 = 44.4% for ChatGPT.

Interpretation: ChatGPT performs well and has a lower unsupported percentage, although more unsupported claims in absolute terms. Relevance should reward sufficient coverage, not require every implementation detail in the prepared answer or reward verbosity automatically.

Q20 - The denominator changes the comparison

Exact question: "What does a variable's data type specify about its values, and what does the cardinality of that type measure?"

Both explain allowed values, operations and cardinality. ChatGPT adds Boolean and 32-bit integer examples; CurriculumAI is more concise. Both receive accuracy and relevance 2.

Observation: Each has exactly one unsupported claim, the cardinality definition. The recorded rates differ: 1/3 = 33.3% versus 1/5 = 20.0%. The variables lecture supports value domains and its slide 28 supports the Boolean values and the integer-range count by inference.

Interpretation: ChatGPT's lower rate reflects more supported material in the denominator, not fewer unsupported assertions. This is an important counterexample to any simple claim that longer answers necessarily have worse percentages.

Q21 - An actual factual qualification

Exact question: "What is an enumerated data type?"

CurriculumAI gives the finite-named-alternatives definition and Scala 3 syntax. ChatGPT gives the general definition, but its C-style example says: "The variable today can only represent one of the seven predefined days." The evidence notes identify the screenshot's language label as C.

Observation: CurriculumAI receives accuracy 2 and ChatGPT 1; both receive relevance 2 and unsupported rates of 100%. The C restriction is materially too strong: the enumerated type's integer-compatible representation and arithmetic-assignment rules do not restrict its values to named enumerators. This concern is consistent with WG14 N1570, §§6.2.5, 6.7.2.2 and 6.5.16.1.

Interpretation: This is a factual issue, rather than a deduction for using C syntax. Both answers remain outside the fixed corpus, demonstrating why equal unsupported rates do not imply equal correctness.

Q23 - Equivalent advanced conceptual performance

Exact question: "What is a union type in Scala?"

Both explain values belonging to alternative types, Scala 3 pipe syntax and type-based handling. CurriculumAI states: "A union type in Scala 3, written A | B, means a value may be either type A or type B." ChatGPT provides a longer explanation, including sealed-trait comparisons.

Observation: Both receive accuracy and relevance 2, no assessable citations and 100% unsupported rates: 5/5 versus 13/13. CurriculumAI has transparency 2 through its external-generation notice; ChatGPT has 0.

Interpretation: The observed content scores do not establish CurriculumAI superiority here. The distinctive feature is disclosure of the answer's external origin. The question is relevant to Scala despite its answer lying outside the prescribed corpus.

5. CurriculumAI Tier-1 and Tier-2 Analysis

Tier separation is essential because the two modes have different evidence conditions. Aggregating them conceals a strong change in corpus support.

For Q01-Q16, the 16 Tier-1 responses have full recorded accuracy and relevance, faithful citations and 0 unsupported claims among 40 assessed. These are results of the saved assessments, not conclusions drawn merely from the presence of citations. They support successful course-bounded answering on this selected set. They do not establish an error-free retriever or universal faithfulness.

For the same Q17-Q24 questions, the comparison is:

Table 5. Matched comparison for Q17-Q24.

System on Q17-Q24	Supported	U/N	Pooled rate	Mean response rate	Accuracy mean	Relevance mean
CurriculumAI Tier 2	13	27/40	67.50%	71.40%	2.000	2.000
ChatGPT	29	56/85	65.88%	66.67%	1.875	2.000

Tier 2 has a slightly higher pooled unsupported rate, by 1.62 percentage points, and a higher mean response rate, by 4.73 points. ChatGPT's accuracy deficit is confined to Q21; relevance is equal throughout these eight questions. Thus Tier 2 does not demonstrate a corpus-support advantage over ChatGPT on matched out-of-corpus questions.

This does not contradict Tier 2's transparency: it explicitly disclaims course grounding. However, disclosure is not claim-level verification and does not turn unsupported statements into supported ones.

The exact routing alignment-Tier 1 for all fully supported questions and Tier 2 for the other eight-is consistent with whole-question evidence-boundary management. It does not independently prove routing correctness, the reason each fallback occurred, or the absence of retrieval failures.

6. Unexpected Findings and Areas of Equivalent Performance

The overall rate advantage is concentrated in course-supported questions. CurriculumAI's lower overall rate is not reproduced on the partially supported or unsupported strata. Across matched individual questions, CurriculumAI has a lower rate on 13, ChatGPT on five, and six are equal.

ChatGPT supplies more supported material in absolute terms. It has 86 supported claims versus CurriculumAI's 53, alongside 103 unsupported claims versus 27. A greater supported count does not establish better learning, but it prevents interpreting a lower unsupported percentage as greater total information coverage.

Answer lengths differ substantially. Counting whitespace-delimited units in the recorded answer field, including code/markup and citation markers but excluding the separate source-reference field, CurriculumAI has 924 units in total, mean 38.5 and median 29.5 per answer. ChatGPT has 5,291 units, mean 220.5 and median 210.0. These are approximate text-length measures, not model-token counts. The claim totals are 80 versus 189, averaging 3.33 versus 7.88 claims per response.

Length and support do not have a simple relationship. Q16 illustrates additional unsupported detail, whereas Q20 illustrates denominator dilution by additional supported examples. The dataset does not establish that length causes unsupported claims, nor that concise answers produce better educational outcomes.

Uncited answers can be fully supported. ChatGPT Q01, Q02 and Q04 have 0% unsupported rates despite having no citations. Conversely, a transparent external answer can have 100% unsupported claims, as CurriculumAI Q21, Q23 and Q24 do.

Citation faithfulness has no observed failures here. No cited response is recorded as factually incorrect. This is an observation about 16 selected responses; with no recorded partial or failed citation cases, the dataset offers limited evidence about how the assessment detects citation failures.

7. Critical Interpretation in Relation to RQ2 and RQ3

The research-question wording was checked in §1.6 of the supplied dissertation draft (Ch1-5_RAG_November_version1.docx).

RQ2 compares accuracy, relevance, citation faithfulness and source transparency on controlled conceptual Scala questions. The recorded data favour CurriculumAI in mean accuracy and relevance, with differences confined to two and five questions respectively. It supplies more inspectable evidence and more explicit provenance disclosure. A comparative claim about citation-faithfulness quality is unavailable because ChatGPT supplies no assessable citations. The appropriate finding is successful citation support within CurriculumAI's cited subset and a citation-availability difference between systems.

RQ3 concerns reduction of corpus-unsupported claims, including uncited claims, under the same ground truth and rubric. The saved counts support a descriptive overall reduction from 54.50% to 33.75%. The more precise interpretation is that the benefit is concentrated in the 16 course-supported Tier-1 questions. There is no observed matched-question reduction for Tier 2; ChatGPT's pooled rates are slightly lower in both remaining question groups.

The findings therefore support the project's bounded evidence-management aim more strongly than a claim of uniformly superior general answering. Model differences, unequal access to the corpus, question ambiguity and judgement-sensitive scoring prevent attributing the full observed difference solely to retrieval. No student-learning improvement or causal architectural effect is established.

8. Validity and Methodological Limitations

Small purposive dataset. Twenty-four selected questions are not a representative random sample of Scala questions or student needs. The 16/4/4 composition weights the overall comparison towards the course-supported group. Q14 overlaps with Q19's class component, and Q20 overlaps with Q22's cardinality definition, as the ground-truth notes acknowledge.

One response per condition. One retained first response per question avoids selective regeneration but does not estimate run-to-run variability. Paired questions improve comparability, while repeated topics and claims within answers limit independence. Claims should not be treated as 269 independent experimental observations for significance testing.

Unequal system conditions. CurriculumAI has course evidence; ChatGPT was not supplied that corpus. This is a comparison of deployed system configurations, not a controlled ablation isolating RAG.

Corpus definition is inconsistent across records. The guide specifies selected lecture slides only, whereas the completed unsupported-claim assessment uses the whole ZIP, including DOCX exercises and the nested Scala example archive. Q02 also displays a DOCX citation.

Claim segmentation involves judgement. Compound statements, examples, paraphrases and repeated explanations can be split differently. Q15's single Java-qualified claim is particularly consequential. Q10's mirror analogy is counted, whereas some other analogies or generic benefits are excluded or grouped. Correlated definition/example claims may be counted separately. These choices affect both numerator and denominator.

Corpus support permits interpretation. Explicit statements, worked examples and clear inference are accepted. Boundary cases include Q02's constant inference, Q03's example, Q20's integer-range count and Q22's distinction between unbounded mathematical integers and Scala's finite Int. External factual references cannot retrospectively become course evidence.

Relevance criteria were refined after collection. A curriculum-context interpretation is legitimate for the intended task, but post-collection refinement introduces researcher degrees of freedom. The guide still states a broad question-coverage definition.

Short questions can be ambiguous. Q14 lacks an explicit programming qualifier, and several collection questions omit Scala. CurriculumAI's application context may reduce ambiguity. This is an observed system-level benefit candidate, but it also limits causal comparisons against a general-purpose chatbot without equivalent contextual instructions.

Technical correctness has contextual and version boundaries. Q05's zero is about failure to answer Scala for … yield; it should not be reported as every Python statement being false. Q21 identifies a specific C semantics problem. Q13's private[this] discussion and other version-sensitive explanations deserve a declared Scala-version scope.

Metric scales have limits. Means of three-point ordinal scores are descriptive summaries, not interval-scale proof of effect size. Ceiling scores for all CurriculumAI accuracy/relevance entries limit discrimination. A pooled unsupported rate weights claims, while a response mean weights questions; neither is a factual-error rate or direct pedagogical-quality measure.